

Geospatial ML for Solar

A curated, verified reading list — to learn the domain properly and fill the gaps between architecture, ML, and GeoAI.

Companion to the *Urban Rooftop Solar Potential* project · Aug 2026 · every link verified against primary sources · old foundations + current (2023–25) work

How to use this. You don't need all of it. Skim §1 to close CRS/raster/vector gaps, get hands-on with §2, then go deep on §3–4 (the actual method) and §5 (Esri's world, for the interview). §6 is a lookup table you'll return to while building. §7 is for framing the "why geospatial / why Esri" story.

Suggested path (≈2–3 weeks of evenings): Week 1 — §1 fundamentals + §2 GeoPandas/rasterio hands-on → Week 2 — §4 solar methodology (Sunroof + NREL/NLR) + §3 U-Net/DeepRoof → Week 3 — §5 ArcGIS API + `arcgis.learn`, alongside building Phase 1.

1 — GIS & Geospatial Fundamentals (close the gaps)

What is GIS? — Esri (current)

The standard framing of what GIS is and does; read once before an Esri interview.

esri.com/en-us/what-is-gis/overview

Coordinate Systems, Map Projections & Transformations — Esri, ArcGIS Pro docs

Canonical geographic-vs-projected CRS explanation — why blind reprojection breaks analysis. Fills the CRS gap directly.

doc.esri.com/en/arcgis-pro/latest/help/mapping/properties/coordinate-systems-and-projections.html

Coordinate Reference Systems — QGIS Documentation v3.44

A second, tool-agnostic take on CRS/projections (UTM, WGS84, on-the-fly projection) to triangulate the concept.

docs.qgis.org/3.44/en/docs/gentle_gis_introduction/coordinate_reference_systems.html

The Nature of Geographic Information — DiBiase et al., Penn State (open textbook)

Free open-access GIS&T textbook (raster/vector, data models, remote sensing) built for self-study.

courses.ems.psu.edu/natureofgeoinfo/

Intro to Geospatial Raster & Vector Data with Python — The Carpentries Incubator

Hands-on, Python-first bridge from GIS concepts to code (geopandas, rioxarray) — ideal on-ramp for a programmer new to raster/vector.

carpentries-incubator.github.io/geospatial-python/

Geographic Information Science and Systems, 4th ed. — Longley, Goodchild, Maguire & Rhind (Wiley)

The standard graduate GIS reference for the theory Esri's material assumes. Reference, not cover-to-cover.

wiley.com — ISBN 9781119128458

2 — Python Geospatial Stack (the tools you'll actually type)

GeoPandas — Getting Started — GeoPandas project

Official docs for the GeoDataFrame — used constantly for footprints, parcels, spatial joins, aggregation.

geopandas.org/en/stable/getting_started/introduction.html

GeoPandas Basics: Maps, Projections & Spatial Joins — Real Python

Strong practical tutorial for the CRS + spatial-join workflow (join roof results to tracts).

realpython.com/geopandas/

Shapely Documentation (v2.x) — Shapely project

The geometry engine under GeoPandas (polygons, buffers, intersections) — roof-plane geometry ops.

shapely.readthedocs.io/en/stable/

Rasterio Documentation — Rasterio / Sean Gillies

Read/write rasters (DSM, NAIP, Sentinel-2) — the backbone of the raster side.

rasterio.readthedocs.io/en/stable/

Reading Raster Files with Rasterio — Automating GIS Processes, Univ. of Helsinki

Applied tutorial for raster I/O, clipping, band math.

autogis-site.readthedocs.io/en/latest/lessons/lesson-7/overview.html

rioxarray Documentation — corteva

The xarray + rasterio bridge — needed once you handle multi-band / multi-temporal stacks (Sentinel-2 time series).

corteva.github.io/rioxarray/stable/

PDAL — Point Data Abstraction Library — PDAL project

The standard for LiDAR/point-cloud pipelines (classification, DSM/DTM generation) — core to roof-plane extraction.

pdal.io

laspy Documentation — laspy project

Lighter pure-Python alternative to PDAL for reading/filtering LAS/LAZ.

laspy.readthedocs.io/en/latest/

OSMnx — docs + building-footprints example — Geoff Boeing

Pull footprints / street networks from OpenStreetMap for urban context.

osmnx.readthedocs.io

3 — Remote Sensing + Deep Learning for Buildings/Rooftops

U-Net: Convolutional Networks for Biomedical Image Segmentation — Ronneberger, Fischer & Brox (2015)

The encoder-decoder architecture under nearly every modern building/roof segmentation pipeline. Seminal.

arxiv.org/abs/1505.04597

Mask R-CNN — He, Gkioxari, Dollár & Girshick, ICCV (2017)

Instance segmentation for per-building / per-roof-plane masks, not just building/non-building pixels.

arxiv.org/abs/1703.06870

SpaceNet — CosmiQ Works / Maxar / IEEE GRSS (since 2016)

The standard open benchmark for building-footprint extraction from satellite imagery.

spacenet.ai

Global ML Building Footprints (repo + method) — Microsoft (2023–24, ~1.4B buildings)

The segmentation → polygonization pipeline behind Microsoft's open footprints — a directly usable blueprint.

github.com/microsoft/GlobalMLBuildingFootprints

Continental-Scale Building Detection from High-Res Satellite Imagery — Sirko et al., Google Research (2021)

The methodology behind Google Open Buildings (U-Net at 50 cm) — the India-relevant footprint source.

arxiv.org/abs/2107.12283

DeepRoof: Solar Potential Estimation Using Rooftop Imagery — Lee, Iyengar, Feng, Shenoy & Maji, KDD '19

Extracts roof planar segments/orientation from imagery and estimates solar potential without LiDAR — key for the India (no-LiDAR) fallback.

dl.acm.org/doi/10.1145/3292500.3330741 (open mirror on ResearchGate)

SolarNet+: city-scale rooftop PV considering roof superstructures — Li, Krapf, Mou, Shi & Zhu, Applied Energy (2024)

Current method modeling orientation + superstructures (chimneys/vents) that cut usable area — closest to your ML step.

github.com/lqycrystal/SolarNet_plus

4 — Solar Potential Estimation Methodology (the heart of it)

Project Sunroof — Methodology & Inputs — Google

Official description of turning imagery + LiDAR DEMs into per-roof-segment potential (CNN scoring, planar segmentation, panel placement). Your closest whole-pipeline precedent.

google.com/get/sunroof/assets/data-explorer-methodology.pdf

Rooftop Solar PV Technical Potential in the US: A Detailed Assessment — Gagnon, Margolis, Melius, Phillips & Elmore, NREL/NLR (2016), TP-6A20-65298

The canonical US rooftop technical-potential study; full GIS+LiDAR+PV pipeline at national scale. Your quantitative benchmark.

docs.nrel.gov/docs/fy16osti/65298.pdf (NREL renamed NLR, Dec 2025)

Using GIS-based methods & lidar to estimate rooftop solar technical potential in US cities —

Margolis et al., Environ. Res. Lett. 12(7) (2017)

Peer-reviewed city-scale companion method — closest published analogue to your project scope.

iopscience.iop.org/article/10.1088/1748-9326/aa7225

Estimating rooftop solar potential across the US using GIS, lidar & statistical modeling —

Gagnon et al., Environ. Res. Lett. 13(2) (2018)

Extends the lidar method statistically to areas lacking lidar — directly relevant to your India degradation problem.

iopscience.iop.org/article/10.1088/1748-9326/aaa554

PVWatts — API & Technical Manual — NREL/NLR (API ongoing; manual Dobos, 2014)

The standard citable model converting roof geometry + irradiance → expected energy yield.

developer.nrel.gov/docs/solar/pvwatts/

PVGIS — Background & Methodology — European Commission JRC (PVGIS 5)

Free satellite-derived irradiance + validated PV model — the methodological comparison to PVWatts and your India irradiance source.

joint-research-centre.ec.europa.eu/photovoltaic-geographical-information-system-pvgis

r.sun — GRASS GIS Reference Manual — GRASS Development Team

The open-source solar irradiance/irradiation raster model (beam, diffuse, reflected, terrain shadowing) — your recommended surface-radiation engine.

grass.osgeo.org/grass-stable/manuals/r.sun.html

Solar Radiation Analysis Tools (Area / Raster Solar Radiation) — Esri, ArcGIS Pro docs

Esri's own insolation model — the "Esri way." Note legacy Area is being superseded by Raster Solar Radiation.

doc.esri.com/en/arcgis-pro/latest/tool-reference/spatial-analyst/an-overview-of-the-solar-radiation-tools.html

Urban-Scale Rooftop PV Potential Using Open-Source Software & Public GIS Datasets —

Smart Cities 7(6):153 (2024, MDPI)

A recent, open-source, public-data city-scale workflow — almost a blueprint for this exact portfolio project.

doi.org/10.3390/smartcities7060153

Disparities in rooftop PV deployment in the US by race & ethnicity — Sunter, Castellanos & Kammen, Nature Sustainability (2019)

The foundational solar-equity paper — grounds your census-tract equity overlay in real literature.

nature.com/articles/s41893-018-0204-z

5 — Esri / ArcGIS Ecosystem for GeoAI (read before the interview)

ArcGIS API for Python — Documentation — Esri

The core Python library for mapping/analysis/publishing — essential fluency for an Esri DS/ML role.

developers.arcgis.com/python/latest/

Geospatial Deep Learning with arcgis.learn — Esri Developer Guide

Official guide to `arcgis.learn` (object detection, pixel classification, point-cloud models on PyTorch/fastai) — your ML step, the Esri way.

developers.arcgis.com/python/guide/geospatial-deep-learning/

Deep Learning in ArcGIS Pro — Esri, ArcGIS Pro docs

The Image Analyst DL workflow (export training data → train → inference) inside ArcGIS Pro.

pro.arcgis.com/en/pro-app/latest/help/analysis/deep-learning/deep-learning-in-arcgis-pro.htm

GeoAI (toolset overview) — Esri, ArcGIS Pro docs

Esri's own framing of where AI/ML sits across its toolsets — speak their vocabulary in interviews.

doc.esri.com/en/arcgis-pro/latest/help/analysis/ai/geoai.html

Area Solar Radiation (Spatial Analyst) — Esri, ArcGIS Pro tool reference

Esri's native insolation tool — the direct counterpart to any solar method you build independently.

pro.arcgis.com/en/pro-app/latest/tool-reference/spatial-analyst/area-solar-radiation.htm

ArcGIS Living Atlas of the World — Esri

Curated global layers (imagery, demographics, environment) — both a data source and a flagship product to know.

livingatlas.arcgis.com

ArcGIS Online — Product Overview — Esri

The SaaS GIS platform where you'll publish and host your deliverable.

esri.com/en-us/arcgis/products/arcgis-online/overview

What Is GeoAI? — Esri Capabilities

Esri's one-page positioning of GeoAI as fused AI + spatial data science — mirror this language in your application.

esri.com/en-us/capabilities/geoai/overview

6 — Datasets & APIs Reference (the lookup table)

NSRDB — National Solar Radiation Database — NLR (formerly NREL)

Satellite-derived irradiance (GHI/DNI/DHI) + weather; strong US, some international. Free, API key.

developer.nlr.gov/docs/solar/nsrdb/

PVWatts API (v8) — NLR (formerly NREL)

PV energy production from system specs + weather. Free, API key.

developer.nlr.gov/docs/solar/pvwatts/

PVGIS — EC Joint Research Centre

Irradiance + PV simulation; SARA covers most of Asia incl. India, ERA5 fully global. Free, no key.

re.jrc.ec.europa.eu/pvg_tools/en/

USGS 3DEP / The National Map — USGS

High-res LiDAR + derived DEMs, ~99% US coverage. US only. Free, no key.

usgs.gov/3d-elevation-program

NAIP — USDA

Orthorectified aerial imagery, 0.3–1 m. CONUS only. Free (also via Planetary Computer / Earth Engine).

fsa.usda.gov/conservation-programs/naip

Microsoft Global ML Building Footprints — Microsoft

1.4B+ AI footprints, some with heights; global incl. substantial India. Free, CDLA-Permissive-2.0.

github.com/microsoft/GlobalMLBuildingFootprints

Google Open Buildings — Google Research

1.8B AI footprints w/ confidence; Africa, South & SE Asia (incl. India), LatAm. Not US/Europe. Free, CC-BY-4.0/ODbL.

sites.research.google/gr/open-buildings/

OpenStreetMap via osmnx — OSM / Geoff Boeing

Street networks, buildings, POIs; global, quality varies. Free, MIT.

osmnx.readthedocs.io

Overture Maps Foundation — Linux Foundation

Open footprints, places, transport, admin boundaries with stable IDs; global. Free/open.

overturemaps.org

Sentinel-2 via Microsoft Planetary Computer — Microsoft / ESA

Global L2A archive (2016–, 13 bands, 10–60 m) via STAC. Global. Free (signed-URL token). Your India imagery.
planetarycomputer.microsoft.com/dataset/sentinel-2-l2a

NASA POWER API — NASA

Global solar/meteorological data; confirmed India coverage. Free, no key. Your India irradiance.
power.larc.nasa.gov

7 — **Domain & Career Framing** (the "why geospatial / why Esri" story)

What Is GeoAI? — Esri

Esri's canonical positioning of GeoAI — vocabulary worth mirroring in your application.
esri.com/en-us/capabilities/geoai/overview

Spatial Analytics & Data Science — Esri Capabilities

How Esri positions "spatial data science" as distinct from GeoAI — shows how they segment the space you're entering.
esri.com/en-us/capabilities/spatial-analytics-data-science/overview

Spatial Data Science: With Applications in R — Pebesma & Bivand (free online, updated 2025)

The standard academic reference framing spatial-data-science fundamentals (concepts transfer beyond R).
r-spatial.org/book/

* **India-Version Data Notes**

For when you build Phase 3. The honest headline: **footprints and irradiance are well-served; high-resolution elevation is the real gap.**

Irradiance: NASA POWER (global, no key) and PVGIS (SARAH covers India; ERA5 global fallback) are the best India-covering equivalents to NSRDB/PVWatts. Pair with `pvlib` for the PV model.

Footprints: Google Open Buildings treats India as a priority region; Microsoft ML Footprints also has substantial India coverage. Comparatively well-served.

Geoportal / DEM: ISRO Bhuvan (national geoportal) and CartoDEM (~30 m, from Cartosat-1) are the national analogues — far coarser than US 3DEP.

The key gap — LiDAR: there is *no* Indian public program comparable to USGS 3DEP's free sub-1 m LiDAR. This is the single biggest constraint for shading-aware rooftop analysis in India — expect to fall back to coarser CartoDEM/Copernicus DEM, commercial/drone LiDAR, or DEM-free imagery + footprint + shadow methods (this is exactly where *DeepRoof*, §3, earns its place).

Every link above was verified to resolve against a primary source (Aug 2026). A couple of Esri GeoAI case-study blog posts were dropped rather than cited because automated verification was bot-blocked — §5 already covers the ecosystem with eight confirmed sources. Companion document: *Urban Rooftop Solar Potential — Project Spec*.